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How the Chief Financial Officer (CFO) affects digital transformation

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How Chief Financial Officer (CFO) affects digital transformation

Abstract: This paper examines the influence of Chief Financial Officer (CFO) characteristics on digital transformation. Using manufacturing firms listed on the A-share market of the Shanghai and Shenzhen stock exchanges between 2007 and 2023, the research results demonstrate that CFO characteristics significantly impact digital transformation. Specifically, male CFOs and CFOs with higher levels of education have a positive impact on digital transformation, while older CFOs have a negative impact. Furthermore, the effects of CFO characteristics on digital transformation differ between different kinds of firms. This paper makes a significant contribution to the fields of digital transformation and upper echelons theory, and offers valuable guidance to firms seeking to achieve digitization.

Key words: digital transformation; chief financial officer; CFO gender; CFO age; CFO education

1. Introduction

In the digital economy, achieving digital transformation has become a key concern for academics and practitioners alike (Hanelt et al., 2021; Zhai et al., 2022; Yao et al., 2024). Within an organization, top executives are usually responsible for digital transformation (Zhu et al., 2022). Therefore, substantial evidence from existing studies supports the claim that top managers have a significant impact on firms' digital transformation.

The literature validates the significant contributions of both the CEO (Liu and Wu, 2024) and the CTO (Firk et al., 2021) to digital transformation. However, the impact of CFOs on digital transformation remains unclear (Yao et al., 2024). In practice, digital transformation is a two-step process: it begins with the CEO's decision to implement it, and ends with the CFO's investment. Furthermore, it has been suggested that the role of the CFO has undergone a significant change, evolving from a primarily financial and accounting role to one that plays an instrumental part in defining and guiding the firm's strategic direction (Uhde et al., 2017). During digital transformation, CFOs can provide valuable financial insights into the costs and benefits of digital initiatives, assess the financial risks associated with digital investments, and help CEOs make informed decisions about allocating resources. Therefore, it is highly likely that a firm's CFO exerts a significant influence on digital transformation.

In light of the above, a pivotal research question is to explore the potential influence of CFOs on digital transformation. According to the upper echelons theory, top executives do not operate in isolation; rather, their characteristics, such as demographics, filter the complex and often ambiguous information they receive from the business environment. Therefore, CFOs' characteristics can shape their perception of the business environment, their evaluation of opportunities and risks, and their decision-making (Hambrick, 2007). In the context of digital transformation, the upper echelons theory provides a valuable lens through which to investigate the impact of CFO

characteristics. Specifically, we examine the influence of three key CFO characteristics — gender, age and education — on digital transformation. These three characteristics have been selected for investigation for several reasons. Firstly, they can be measured quantitatively, allowing for a more objective analysis. Secondly, these characteristics may be associated with variations observed in managerial decision-making (Ginesti et al., 2021). For example, gender and age may affect an executive's willingness to take risks, while education can provide the necessary knowledge and skills to understand and implement digital transformation strategies. By examining these characteristics, we aim to gain a deeper understanding of how CFOs' characteristics can influence their firms' digital transformation.

This paper makes three key contributions to the existing body of literature. Firstly, it makes a contribution to the literature on the drivers of digital transformation by demonstrating the multifaceted effect of CFO characteristics on a firm's digital transformation. Secondly, it makes a contribution to the literature on the impact of CFOs by establishing a clear link between their characteristics and digital transformation. Thirdly, it advances the current understanding of the effects of CFO characteristics by identifying the relevant contingency factors. This provides a more nuanced and in-depth explanation of when and how firms benefit from their CFOs in digital transformation.

This paper is structured as follows. Section 2 reviews the existing literature and presents a set of hypotheses. Section 3 outlines the research method. Section 4 describes the research results and offers a series of robustness checks and additional analyzes. Section 5 discusses the theoretical and practical implications of the research, as well as limitations for future research.

2. Literature review and research hypotheses

2.1 Top executives and digital transformation

In recent years, there has been a notable increase in the number of studies on digital transformation (Verhoef et al., 2021), with most scholars acknowledging its strategic importance (Kraus et al., 2022). A large body of in-depth research has consistently emphasized the pivotal role of top executives in ensuring digital transformation (Zhu et al., 2022).

In light of this, a growing body of research has examined the relationship between top executive characteristics and digital transformation. This line of research is aligned with the upper echelons theory. According to this theory, organizational outcomes, including digital transformation, directly reflect the values and cognitive biases inherent in top executives' characteristics (Bromiley and Rau, 2016; Neely et al., 2020). Studies based on the upper echelons theory have identified several key factors when examining these top executive characteristics. Notably, the gender,

age and education of top executives have emerged as significant determinants. These characteristics are believed to have a profound impact on a top executive's values and cognitive biases (Wang et al., 2016; Wille et al., 2018).

However, although a substantial amount of scholarly attention has been given to the characteristics of CEOs and CTOs in driving digital transformation, there has been a relative lack of in-depth studies examining the role of CFOs. This is somewhat surprising, given the significant changes to the CFO role (Ojeka et al., 2019). In light of this, the present paper aims to explore how the gender, age and education of CFOs impact digital transformation. By doing so, the paper seeks to contribute to a more holistic understanding of the role of top executives in digital transformation and provide valuable insights for firms looking to leverage their CFOs' expertise in this critical area.

2.2 Hypothesis development

The existence of gender differences in risk attitudes and related behaviors is well documented (Klenke, 2003). Most previous studies have supported the idea that women are more risk-averse than men (Bao et al., 2014) and that these different risk attitudes influence their decision-making (Khan and Vieito, 2013). It has also been demonstrated that women are more inclined to adopt strategies that minimize potential adverse outcomes and provide greater security. For example, the portfolios of female investors have been found to be less risky than those of their male counterparts (Luo and Salterio, 2022). Female investors tend to consider risk attributes more deeply, such as the possibility of loss and ambiguity. They prioritize reducing the risk in their portfolio investments, carefully weighing up the potential downsides before making decisions. This cautious approach is rooted in their risk-averse nature. Furthermore, a growing body of evidence suggests that these gender-specific risk-taking preferences also manifest in firm decision-making at top executives level. For example, empirical evidence suggests that firms with female CFOs tend to adopt less risky policies and actions (Schopohl et al., 2021). Croson and Gneezy (2009) propose three potential explanations for the observed gender differences in attitudes towards risk. Firstly, research suggests that women experience greater fear and nervousness than men when faced with uncertainty. Secondly, evidence suggests that women display lower levels of confidence and competence than men in a crisis. Thirdly, studies have shown that women are more likely to perceive risky situations as threats rather than challenges.

The digital transformation of a firm is highly risky. It requires substantial investment in new technologies, as well as changes to business processes and organizational culture. Each of these elements carries risks, including technological failures, employee resistance and failing to achieve the desired return on investment. If female CFOs are indeed more risk-averse and this is reflected in their decision-making, it is reasonable to anticipate lower levels of

digital transformation for firms led by female CFOs. They may be more reluctant to allocate resources to digital transformation due to concerns about potential risks and uncertainties. This suggests a potential negative correlation between female CFOs and the extent of a firm's digitization. In light of these considerations, we propose the following hypothesis:

H1: Firms with male CFOs have higher digital transformation than firms with female CFOs.

Scholars generally agree that age is a significant characteristic of top executives, given that managerial actions are associated with shifts in individual orientation that accompany ageing (Li et al., 2017). This suggests that, as they grow older, top executives tend to favour maintaining the status quo within the organization. They often prefer a quiet and stable professional life. This preference for stability is closely tied to increased risk aversion (Serfling, 2014). Digital transformation inevitably involves considerable risks and substantial additional costs (Verhoef et al., 2021). Therefore, older CFOs may be reluctant to advocate digital transformation.

Firstly, as CFOs age, they are likely to experience a natural decline in both mental and physical energy. Digital transformation is a long-term investment project demanding a high level of cognitive engagement. It requires CFOs to understand complex technological concepts, analyze vast amounts of data related to digital initiatives and make informed decisions about resource allocation. With reduced mental and physical energy, older CFOs may find it challenging to sustain the level of commitment and focus required for such a demanding endeavour. Secondly, a preference for a quiet life and a commitment to organizational status quo tend to increase with age. Older CFOs have often spent a significant portion of their careers building and maintaining the existing business model. Digital transformation, by its very nature, is disruptive. It has the potential to alter business operations in fundamental ways. This disruption can create instability, which goes against the nature of older CFOs. Therefore, they may be reluctant to increase investment in digital transformation. Thirdly, older CFOs tend to prioritize financial security above all else. The costs of digital transformation can put a strain on a firm's financial resources, particularly if the return on investment is not immediate or as high as anticipated. Given these financial risks, older CFOs may be more inclined to opt for safer investment options that offer a more stable and predictable return than taking a chance on digital transformation. Fourthly, older CFOs were probably educated at a time when digital transformation was less prevalent, or even absent, from the business landscape. Their early career experiences may have centred more on traditional financial management than on emerging digital technologies. Therefore, they may be unfamiliar with the potential advantages and disadvantages of digital transformation. Without an in-depth understanding of how digital

technologies can generate new business opportunities, they may perceive digital transformation as an unnecessary and risky endeavour rather than a strategic imperative for the firm's future success. Based on the above arguments, we propose the following hypothesis:

H2: There is a negative relationship between the CFO age and digital transformation.

In addition to gender and age, the education of top executives is another characteristic of senior managers that is frequently examined in the upper echelons theory (White and Borgholthaus, 2022). This theory postulates that top executives' education reflects their professional knowledge, which subsequently influences their understanding and analytical abilities.

On the one hand, highly educated CFOs tend to be curious and open to new ideas. With their forward-thinking mindset, they are more likely to embrace the concept of digital transformation and advocate increased investment in this area. They recognize that the long-term benefits of digital transformation, such as improved operational efficiency, an enhanced customer experience and the creation of new business models, far outweigh the short-term costs and risks. Given these long-term advantages, highly educated CFOs are not deterred by the short-term costs and risks associated with the process, such as the initial investment in technology infrastructure and employee training, or potential disruptions to existing operations. Instead, they take a long-term view, recognizing that these short-term challenges are a necessary investment in the firm's future competitiveness. On the other hand, the education of CFOs directly impacts a firm's financial management (Nguyen and Van Nguyen, 2024). A CFO's qualifications are demonstrated not only through a professional degree, but also through their ability to manage risks and costs effectively. Digital transformation is a complex and multifaceted process involving a series of intricate tasks. CFOs with higher levels of education are better equipped to navigate these challenges. Their in-depth knowledge of financial principles, such as cost-benefit analysis, capital budgeting and risk assessment, enables them to accurately evaluate the potential risks and rewards of digital transformation. They can develop comprehensive financial models that consider various scenarios and uncertainties, helping the firm to make more informed digital transformation investment decisions. Additionally, highly educated CFOs often have a broader network of professional contacts and a deeper understanding of industry trends. This network can provide valuable insights and access to resources that facilitate digital transformation. For instance, they may collaborate with technology experts, consultants, and firms that have successfully implemented digital transformation. Their knowledge of industry trends enables them to anticipate future

changes and position the firm accordingly, ensuring it remains at the forefront of digital transformation. Based on the above arguments, the following hypothesis is proposed:

H3: There is a positive relationship between the CFO education and digital transformation.

The theoretical model of this paper is shown in Figure 1.

Insert Figure 1 near here

3 Research method

3.1 Samples

This paper focuses on manufacturing firms listed on the Chinese A-share market of the Shanghai and Shenzhen stock exchanges. There are several reasons why this paper focuses on manufacturing firms. Firstly, the manufacturing industry is a key pillar of the national economy. Therefore, enhancing the competitiveness of manufacturing firms is a key objective in achieving high-quality economic growth. Secondly, the manufacturing industry is a key area for the adoption of digital technologies. In the digital era, manufacturing firms must enhance their competitiveness by capitalizing on the potential of digital transformation (Tian et al., 2023). Thirdly, many Chinese manufacturing firms have already achieved varying degrees of digitization, providing a substantial sample pool for this study (Li et al., 2022).

The data were obtained from the China Stock Market and Accounting Research (CSMAR) database (<https://data.csmar.com/>), one of the most frequently used business and management databases in China. The following steps were employed to collect and match the data. Firstly, the digital transformation data provided by CSMAR was retrieved. Secondly, data on other variables was retrieved from CSMAR. Finally, the digital transformation data was merged with the other variable data, using stock codes and year as identifiers.

This paper used the time frame of 2007 to 2023. We chose 2007 as the baseline year because CSMAR began collecting data on digital transformation in that year. However, the time frame of 2007-2023 is very long. The rapid advancements in digital technologies have reshaped the landscape of digital transformation, thus changing its nature significantly. Therefore, we incorporated year fixed effects into our model to control for the time-related trends and period-specific shifts.

Following the previous research (Liu et al., 2023; Wen et al., 2022), the sample was subjected to a further screening process. Firstly, we excluded companies with special treatment (ST/*ST), as these companies are at risk of

being delisted Secondly, we further excluded firm-year observations with incomplete information. In the end, the final data set for this paper consisted of 14929 firm-year observations, representing 2765 firms.

Sampling bias is a prevalent challenge in business and management research. It often arises when observation units make decisions that result in a subset of a particular population not being observed. As this paper encompasses all manufacturing firms, the sample in this paper is not subject to any constraints in terms of CFOs and digital transformation.

3.2 Measurement

3.2.1 Dependent variable

The dependent variable in this paper is digital transformation. As a long-term, systematic process, it is typically documented in a firm's annual report (Liu et al., 2023). The vocabulary used in a firm's annual report can reflect its strategic direction to a large extent. Therefore, digital transformation can be described by counting the frequency of keywords related to it in a firm's annual reports.

However, it is important to recognize the limitations of this approach. Although it can provide an indication of a firm's strategic intent regarding digital transformation, keyword frequency analysis primarily captures a firm's stated goals and aspirations rather than its actual implementation or investment in digital transformation. For example, a high frequency of related keywords in an annual report may suggest that a firm is committed to digital transformation in principle, but this does not necessarily indicate that it has taken concrete steps to implement it or allocate resources to it. Therefore, researchers must exercise caution when interpreting the results of keyword frequency analysis.

Considering the CSMAR database and following the previous research (Guo et al., 2023; Tian et al., 2022), the keywords related to digital transformation cover both the use of core digital technologies (artificial intelligence technology, blockchain technology, cloud computing technology and big data technology) and the applications of digital technology of firms. The level of a firm's digital transformation was reflected by the frequency of these keywords (see Appendix Table 1). To overcome the potential problem of the data being skewed to the right, we took the natural log of the total number of keywords relating to digital transformation plus one.

3.2.2 Independent variables

The independent variables in this paper are the gender, age and education of the CFO. In accordance with the previous research (Wei et al., 2018), the gender of the CFO was determined by a binary variable, with a value of 1 assigned to males and 0 to females. CFO age was quantified in years. CFO education was measured by an ordinal

variable indicating the level of education received by the CFO. This variable takes values from 1 to 5, with 1 representing secondary school and below, 2 representing junior college, 3 representing a bachelor's degree, 4 representing a master's degree and 5 representing a doctorate.

3.2.3 Control variables

Referring to the previous research (Du et al., 2024; Li and Shao, 2023), a set of control variables was included at the CEO, board and firm levels.

At the CEO level, we controlled for CEO gender, CEO age and CEO education because these CEO characteristics have been demonstrated to affect firm decision-making (Shen, 2021). In accordance with the previous research (Wei et al., 2018), the gender of the CEO was quantified using a binary variable, with a value of 1 assigned to males and 0 to females. CEO age was quantified in years. CEO education was measured using an ordinal variable indicating the level of education received by the CEO. This variable takes values from 1 to 5, with 1 representing secondary school and below, 2 representing junior college, 3 representing a bachelor's degree, 4 representing a master's degree and 5 representing a doctorate.

At the board level, we controlled for board size, board diversity and board independence, as a substantial body of research has demonstrated that these board characteristics exert a significant influence on a range of firm outcomes (Lu et al., 2022). In line with the previous research (Orazalin and Mahmood, 2021), the number of board members was used to quantify board size. The percentage of female directors in relation to the total number of directors was employed to assess board diversity. The proportion of independent directors within a firm's board was utilized to evaluate board independence.

At the firm level, we controlled for factors including firm size, firm age, firm leverage, business growth and state ownership. The variables of firm size and firm age were included as control variables on the grounds that larger and older firms invariably possess greater resources for the purposes of digital transformation. In accordance with the previous research, the firm size was measured by the natural logarithm of the firm's total assets, while the firm age was measured by the number of years since the firm was established (Chizema et al., 2015). The ratio of total debt to total assets, which serves as a measurement of firm leverage (Ren et al., 2023), was included as a control variable on the grounds that firms with disparate levels of leverage possess varying degrees of slack resources that can be allocated to digital transformation. The growth rate of operating income was employed as a measurement of business growth (Li and Shao, 2023). A higher value indicates a faster growth rate of the firm's business income and a more favourable

market outlook. Given that state-owned firms are accorded preferential status in the procurement of loans and other key inputs (Jia et al., 2019), we controlled for state ownership. State ownership was measured by a dummy variable, which took the value of 1 if the firm was state-owned and 0 otherwise (Zhou et al., 2017).

4 Research results

4.1 Descriptive statistics

The descriptive statistics for the variables of interest in this paper are presented in Table 1. It can be seen that the minimum and maximum values of the digital transformation are 0 and 6.139885 respectively, which indicates that there are significant differences between the firms. The mean CFO gender is 0.6645, indicating that 66.45% of CFOs are male. Furthermore, the distribution of the variables is similar to that observed in the previous research.

Insert Table 1 near here

Table 2 shows the correlation coefficients for each pair of variables used in this paper. As can be seen from the table, the correlation coefficients are all relatively low, suggesting that multi-collinearity is not a significant concern.

Insert Table 2 near here

4.2 Regression analysis results

The results of the regression analysis are presented in Table 3. To address potential concerns about unobservable time trends driving the results, the regression model was constructed using year fixed effects. Additionally, since CFO characteristics affect firms' digital transformation, which may require a lag, this paper applies a one-period lag to the CFO characteristics and control variables. This approach can also mitigate the endogenous concern of reverse causality.

Model 1 examined the influence of CFO gender on digital transformation. The coefficient was found to be significantly positive ($\beta=0.0637$, $p<0.01$), thereby supporting H1. Model 2 investigated the impact of CFO age on digital transformation and found the coefficient to be significantly negative ($\beta=-0.0111$, $p<0.01$) thereby supporting H2. Model 3 examined the effect of the CFO education on digital transformation. The coefficient was found to be significantly positive ($\beta=0.1025$, $p<0.01$), thereby supporting H3. Model 4 is the full model, which confirmed the research results.

Furthermore, the regression analysis results offer intriguing insights into the influence of top executives education on a firm's digital transformation. Specifically, the coefficient for CFO education ($\beta=0.0824$, $p<0.01$) is similar in

magnitude to that for CEO education ($\beta=0.0873$, $p<0.01$). This provides compelling evidence that the education of CFOs and CEOs have a comparable impact on digital transformation. The similarity in the coefficients for CFO and CEO education suggests that both roles are equally important for digital transformation. Rather than suggesting that one top executive's education dominates the other's, this indicates a collaborative and synergistic relationship between the CFO and the CEO. Instead, it suggests a collaborative and synergistic relationship between the two. Firms should therefore pay equal attention to the education qualifications of their CFOs and CEOs when recruiting and developing top-level talent, since both roles are key drivers of digital transformation.

Insert Table 3 near here

4.3 Robustness tests

A series of robustness tests were conducted to ascertain the robustness of the research results. Firstly, although a number of control variables have been included, digital transformation is a complex activity, which requires the inclusion of other control variables. Previous research has demonstrated that the equity share owned by a shareholder can affect a firm's behaviour (Kim et al., 2007). Therefore, the largest shareholder has been added as a control variable to test the robustness of the results. Table 4 shows the corresponding results. The research results remain robust following this robustness test.

Insert Table 4 near here

Secondly, as the data covers a relatively long period of time, a number of events could potentially impact the research results of this paper. In line with the previous research (Xia and Liu, 2022), we selected the first and last ten-year segments of the sample for re-examination of the research hypotheses. Tables 5 and Table 6 present the corresponding results. These results are robust to this robustness test.

Insert Table 5 near here

Insert Table 6 near here

Thirdly, we modified the regression model. The total number of keywords related to digital transformation was used as an indicator of digital transformation. As this measurement is count-based, the Poisson regression model was used for estimation in this paper. The results are presented in Table 7. As can be seen, the robustness results are consistent, suggesting that the research results are robust.

Insert Table 7 near here

Fourthly, digital transformation is a long-term, systematic process, which typically requires a significant time frame for implementation. To accommodate the potential for incomplete observation of digital transformation outcomes after a one-year lag, we extended the independent variables by two years to mitigate the issue of reverse causality. Table 8 shows the corresponding results. The research results are consistent with the results in Table 3.

Insert Table 8 near here

Finally, this paper used the Generalized Method of Moments (GMM) to address the issue of endogeneity. GMM is well-suited to address potential reverse causality problems. One of its key advantages is that it selects instrument variables internally, eliminating the need to provide them separately, as is required in other instrumental variable regression methods. Specifically, this paper applied system GMM. The system GMM technique produces more efficient and consistent estimates, compared to different GMM econometric techniques (Baltagi 2008). Furthermore, the system GMM can also address the omitted variable bias and heteroskedasticity issue and produces the estimate's reliability. In this paper, the one-year lagged dependent variable was incorporated as an independent variable to implement system GMM. The corresponding results are presented in Table 9. Firstly, the p-value of the first-order autocorrelation is significant, which implies that the null hypothesis of non-autocorrelation is rejected for the AR(1) test. Secondly, the p-value of the second-order autocorrelation is not significant, implying the inability to reject the null hypothesis of non-autocorrelation for the AR(2) test. Therefore, there is no problem of serial correlation. Thirdly, the non-significance of the Hansen test statistics implies the non-rejection of the null hypothesis of valid instruments for Hansen's tests. Therefore, there is no over-identification problem in the results. Overall, the research results are partly robust.

Insert Table 9 near here

4.5 Heterogeneity tests

The impact of CFO characteristics varies significantly across different firms. This variability can be attributed to a multitude of influencing factors, the most significant of which are firm size, firm age, and ownership type. Firm size often dictates the scale and complexity of financial operations, thereby influencing the most effective type of financial expertise and leadership style required. Similarly, a firm's age may reflect its maturity and established culture, shaping the CFO's role in driving digital transformation. The ownership of a firm also introduces unique governance structures and stakeholder expectations that can impact how CFO characteristics translate into tangible outcomes. Given these

nuanced differences, a one-size-fits-all approach to analyzing the relationship between CFO characteristics and digital transformation would be overly simplistic and potentially misleading. Therefore, this paper employs a subsample analysis based on firm size, firm age and ownership type to improve understanding of this relationship. In line with previous research (Liu et al., 2024; Lu and Wang, 2018), firms with total assets exceeding the median are classified as large, while those with lower assets are classified as small. Firms are designated as older or younger based on the median established year for all firms. State-owned firms are distinguished from non-state-owned firms based on whether the ultimate controller is the state. The research results of the heterogeneity analysis are presented in Table 10, Table 11 and Table 12.

Insert Table 10 near here

Insert Table 11 near here

Insert Table 12 near here

In the previous research (Abdi et al., 2022; Medase, 2020), firm size and firm age have been identified as the two principal moderating variables. As the research results in Table 10 show, we find that the difference in the coefficients of CFO gender is statistically significant ($p < 0.05$), suggesting that the positive effect of CFO gender on digital transformation is more substantial for larger firms ($\beta = 0.1158$, $p < 0.01$) relative to smaller firms ($\beta = 0.0015$, $p > 0.1$). It is commonly believed that larger firms have superior financial resources and capabilities. In contrast, CFOs in smaller firms typically find themselves in a much more constrained financial environment. Due to their limited scale of operations and relatively nascent market positions, smaller firms have fewer internal funds available for investment in digital transformation. They also face significant challenges when it comes to accessing external financing. This shortage of financial resources puts CFOs in smaller firms at a significant disadvantage when it comes to driving digital transformation.

As the research results in Table 11 demonstrate, we find that the difference in the coefficients of CFO age is statistically significant ($p < 0.05$), suggesting that the negative effect of CFO age on digital transformation is more substantial for younger firms ($\beta = -0.0149$, $p < 0.01$) relative to older firms ($\beta = -0.0078$, $p < 0.01$). The liability of newness hypothesis posits that younger firms are at a distinct disadvantage in comparison to their older counterparts. This is primarily because younger firms typically lack experience, have not yet established extensive external industry and business ecosystem ties, and may struggle to gain legitimacy with stakeholders such as customers, investors and

regulators. Given these shortcomings, the obstacles to digital transformation may be exacerbated for younger firms with older CFOs, making the negative influence of CFO age more evident than for older firms.

In recent years, the Chinese government has prioritized digital transformation. Therefore, CFOs in state-owned firms are under increased pressure to implement digital transformation in order to achieve the government's stated goal. Table 12 presents the research results. We find that the difference in the CFO age coefficients is statistically significant ($p < 0.05$), suggesting that the negative effect of CFO age on digital transformation is more substantial for non-state-owned firms ($\beta = -0.0132$, $p < 0.01$) than for state-owned firms ($\beta = -0.0029$, $p > 0.1$). Similarly, the difference in the CFO education coefficients is also statistically significant ($p < 0.05$), indicating that the negative impact of CFO education on digital transformation is greater for non-state-owned firms ($\beta = 0.1454$, $p < 0.01$) than for state-owned firms ($\beta = -0.0170$, $p > 0.1$).

5 Conclusion

5.1 Theoretical implications

In light of the research results, this paper makes three significant contributions to the existing body of literature on this topic. Firstly, it makes a contribution to literature on the drivers of digital transformation, clearly demonstrating the significant effect of CFO characteristics on firm digitization. Previous research on factors influencing digital transformation has tended to focus on the top management team (Mirza et al., 2024), with the CEO (Liu and Wu, 2024) and the CTO (Firk et al., 2021) being two prominent examples. However, there has been a conspicuous lack of research examining the role of CFO characteristics in the context of digital transformation. This research gap is surprising, given that upper echelons theory consistently reports that CFOs play a pivotal role in firm strategy and subsequent organizational outcomes. This paper addresses this issue by conducting a detailed examination of the impact of CFO characteristics on digital transformation. The research results reveal that CFO gender, CFO age, and CFO education all significantly impact firm digitization. By highlighting these impacts, the results extend previous research, significantly expanding the understanding of the antecedents of digital transformation. They demonstrate that digital transformation is influenced not only by the CEO's vision or the CTO's technical capabilities, but also by the unique characteristics of the CFO.

Secondly, this paper contributes to existing literature on the impact of CFO characteristics by establishing a direct link between the two. The role of the CFO in strategic decision-making has long been of great interest to scholars and practitioners. While previous studies have examined the influence of CFO characteristics on firm performance (Datta

et al., 2023) and R&D investment intensity (Ginesti et al., 2021), our research indicates that CFO characteristics are also associated with digital transformation. Specifically, male CFOs and CFOs with higher levels of education have a positive impact on digital transformation, while older CFOs have a negative impact. Our paper thus contributes to the broader literature on the effects of CFOs on organizational outcomes. It provides a more comprehensive understanding of how CFO characteristics can influence a firm's strategic direction and success in the digital age.

Thirdly, this paper contributes to the upper echelons theory by identifying the critical boundary conditions of the effect of CFO characteristics. Upper echelons theory has long been a cornerstone in the study of organizational behavior and strategic management, emphasizing that the characteristics of top executives play a pivotal role in shaping organizational outcomes. It has already well-established the critical role of contingency, highlighting the fact that not all top executives influence organizational outcomes equally (Neely et al., 2020). However, previous research in this area has left a significant gap in our understanding of the potential contingencies of CFO characteristics in the context of digital transformation. This paper goes beyond existing literature by demonstrating that the effects of CFO characteristics are contingent on firm type. By identifying the relevant contingency factors, this paper provides a more nuanced explanation of the circumstances in which a firm benefits most from its CFO during digital transformation. Furthermore, this paper can guide CFOs in understanding how they can best contribute to their firms' digital transformation efforts, based on their organizations' characteristics.

5.2 Practical implications

The research results provide firms with crucial guidance on how to achieve digital transformation. They also offer valuable insights to government and regulatory bodies, helping them to formulate relevant policies that foster a conducive environment for digital transformation. These findings are also closely aligned with the objectives of the United Nations Sustainable Development Goals (SDGs), particularly those relating to innovation, inclusive and sustainable economic growth, and reduced inequalities.

Firstly, the research results highlight the significant impact of CFO characteristics as indicators of underlying cognitive frameworks and risk preferences on digital transformation. The research results suggest that firms seeking to implement digital transformation may benefit from prioritizing CFOs who embody this profile: younger, highly educated men. This is because they statistically demonstrate a higher propensity to champion such transformative changes. However, it is important to interpret these research results with nuance. These characteristics markers represent probabilistic tendencies rather than deterministic rules. The absence of these characteristics does not

preclude successful digital transformation. Indeed, older, female or less educated CFOs may possess compensatory strengths, such as extensive industry experience, superior change management skills or robust stakeholder networks, that are also equally critical. Therefore, rather than adopting a rigid hiring checklist, firms should view these characteristics as key variables within a broader strategic equation. The optimal approach is to leverage these characteristics to build complementary CFO teams.

From the governmental and regulatory standpoint, these findings could inform the development of policies that promote diversity within corporate leadership in the context of digital transformation. This is crucial for reducing inequalities. While the research results suggest certain CFO characteristics, there is a significant risk of policies being formulated based solely on these characteristics. For instance, policies that overly emphasize recruiting male CFOs with high educational qualifications may unintentionally marginalize female candidates or those with alternative but equally valuable skills and experience. Similarly, focusing solely on age could result in older CFOs, who have a wealth of industry knowledge, strategic vision and leadership skills that are invaluable for navigating firms through the complexities of digital transformation, being dismissed. Rather than setting rigid criteria based on specific characteristics, policies could incentivize firms to evaluate candidates based on their overall capabilities.

Secondly, the research results delve deeper into the nuanced effects of CFO age, highlighting that its negative impact is not uniform across all kinds of firms. Notably, the negative impact of an older CFO is more pronounced in younger and non-state-owned firms. This finding emphasizes the importance of younger firms and non-state-owned firms exercising particular caution when selecting and retaining CFOs.

For government and regulatory bodies, this insight could inform the development of targeted support and guidance for different kinds of firms. For example, the government and regulatory could offer younger firms targeted training programmes to help them understand the role of CFOs in digital transformation. Additionally, they could implement stricter oversight or provide more detailed guidelines for these firms regarding CFO selection and performance evaluation. This would ensure that they are well-equipped to navigate digital transformation, build resilient infrastructure, and foster innovation, as outlined in the SDGs. By doing so, government and regulatory bodies can play a pivotal role in promoting a more inclusive, innovative and sustainable corporate landscape that aligns with the SDGs.

5.3 Limitations and future research

This paper has its limitations, which provide avenues for future research. Firstly, the paper uses second-hand data to measure digital transformation. Although this method is widely used, it is recognized that it can introduce bias. Therefore, further research could develop a more comprehensive measurement of digital transformation by including additional data sources.

Secondly, the paper focuses on observable CFO characteristics and does not consider how other managerial characteristics may influence digital transformation. Further research could incorporate psychological traits of CFOs to explore their influence on digital transformation.

Thirdly, this paper only partially addresses the issue of endogeneity. Future research could employ alternative strategies, such as dynamic panel estimators, to address the issue of endogeneity more effectively.

Fourthly, the measurement of CFO education may be subject to bias. Our approach assumes that progressing from a bachelor's to a master's degree has the same effect as progressing from a master's to a doctoral degree. Therefore, future research could develop a more effective method of measuring CFO education.

Fifthly, this paper focuses solely on the influence of CFOs on digital transformation, treating CEO characteristics as mere control variables. Further research could examine the dynamic interaction effects between CFOs and CEOs on digital outcomes.

Finally, the sample was limited to listed Chinese manufacturing firms, which may limit the generalizability of the research results. Future studies could replicate this research in different industries and countries to improve its generalizability.

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Appendix Table 1 The keywords of digital transformation

Dimensions of digital transformation	keywords
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Artificial intelligence technology	Machine learning, artificial intelligence, face recognition, business intelligence, identity verification, deep learning, biometrics, image understanding, semantic search, voice recognition, intelligent robotics, intelligent data analysis, autonomous driving, natural voice processing
Blockchain technology	Bitcoin, distributed computing, consensus mechanism, federated chain, decentralization. The terms of cloud computing technology include EB-level storage, multi-party secure computing, brain-like computing, stream computing, green computing, in-memory computing, cognitive computing, converged architecture, graph computing, Internet of Things, information physical systems, billion concurrency, and cloud computing
Cloud computing technology	EB-level storage, multi-party secure computing, brain-like computing, stream computing, green computing, in-memory computing, cognitive computing, converged architecture, graph computing, Internet of Things, information physical systems, billion concurrency, cloud computing
Big data technology	Mixed reality, data visualization, data mining, text mining, virtual reality, heterogeneous data, augmented reality, credit investigation
Digital technology applications	B2B, B2C, C2B, C2C, fintech, NFC payment, O2O, third-party payment, e-commerce, industrial Internet, Internet finance, Internet healthcare, fintech, open banking, quantitative finance, digital finance, digital marketing, netlink, unmanned retail, mobile Internet, mobile payment, smart agriculture, smart wear, smart grid, smart environmental protection, smart home, smart transportation, smart customer service, smart energy, smart investment, smart cultural travel, smart medical, smart marketing

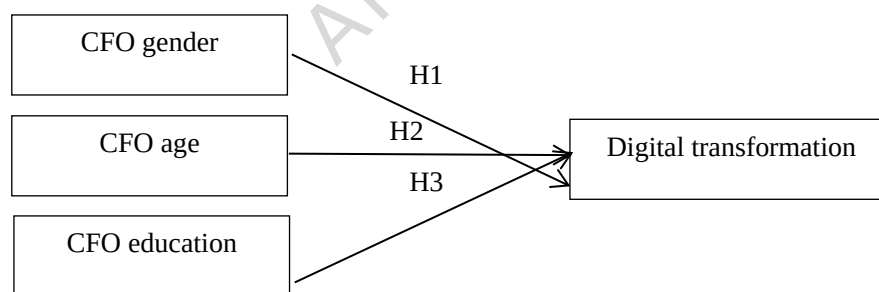


Figure 1 Theoretical model of this paper

Table 1 Descriptive statistics

Variables	Observations	Mean	SD	Min	Max
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Digital transformation	14929	1.1762	1.2691	0	6.1399
CFO gender	14929	0.6645	0.4722	0	1
CFO age	14929	45.8898	6.4737	27	71
CFO education	14929	3.1013	0.7465	1	5
CEO gender	14929	0.9321	0.2515	0	1
CEO age	14929	49.9509	7.1577	26	82
CEO education	14929	3.3331	0.9346	1	5
Board size	14929	8.4043	1.5905	4	17
Board diversity	14929	0.1580	0.1334	0	0.8
Board independence	14929	0.3751	0.0547	0	0.75
Firm size	14929	21.8984	1.1477	17.8789	27.1218
Firm age	14929	17.4290	6.1458	1	63
Firm leverage	14929	0.4001	0.3860	0.0071	18.9398
Business growth	14929	0.3756	9.7142	-0.9818	944.0996
State ownership	14929	0.2503	0.4332	0	1

Table 2 Correlation results

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 Digital transformation	1.0000														
2 CFO gender	-0.0159	1.0000													
3 CFO age	0.0201	-0.0114	1.0000												
4 CFO education	0.1013	0.1169	-0.1754	1.0000											
5 CEO gender	-0.0154	0.0249	0.0374	0.0145	1.0000										
6 CEO age	0.0512	0.0101	0.1754	0.0005	0.0625	1.0000									
7 CEO education	0.0886	0.0418	-0.0096	0.2346	0.0275	-0.1070	1.0000								
8 Board size	-0.1124	0.0530	0.0379	0.0448	0.0857	0.0176	0.0713	1.0000							
9 Board diversity	0.0443	-0.1855	-0.0022	-0.0442	-0.2501	0.0225	-0.0434	-0.1054	1.0000						
10 Board independence	0.0889	-0.0086	-0.0037	0.0039	-0.0581	0.0023	-0.0006	-0.5160	0.0468	1.0000					
11 Firm size	0.1294	0.0562	0.1294	0.2167	0.0202	0.1006	0.1629	0.2231	-0.0986	0.0043	1.0000				
12 Firm age	0.1532	-0.0138	0.1630	0.0596	-0.0342	0.1540	0.0178	-0.0237	0.0628	0.0442	0.1878	1.0000			
13 Firm leverage	-0.0310	0.0501	-0.0092	0.0498	0.0050	-0.0405	0.0243	0.1031	-0.0333	-0.0140	0.1492	0.0470	1.0000		
14 Business growth	0.0000	-0.0143	-0.0049	0.0126	-0.0212	0.0068	0.0148	0.0167	0.0029	-0.0057	0.0118	0.0008	0.0059	1.0000	
15 State ownership	-0.1203	0.1155	0.0736	0.1521	0.0790	0.0193	0.1617	0.3018	-0.2001	-0.0769	0.2862	0.0896	0.1229	-0.0041	1.0000

Table 3 Regressions of CFO characteristics on digital transformation (one year lag)

Variables	Model 1	Model 2	Model 3	Model 4
CFO gender	0.0637*** (0.0239)			0.0482** (0.0239)
CFO age		-0.0111*** (0.0018)		-0.0088*** (0.0018)
CFO education			0.1025*** (0.0154)	0.0824*** (0.0158)
CEO gender	0.0091 (0.0454)	0.0133 (0.0453)	0.0055 (0.0453)	0.0142 (0.0453)
CEO age	-0.0035** (0.0016)	-0.0022 (0.0016)	-0.0034** (0.0016)	-0.0025 (0.0016)
CEO education	0.1006*** (0.0122)	0.1006*** (0.0122)	0.0861*** (0.0124)	0.0873*** (0.0124)
Board size	-0.0076 (0.0087)	-0.0059 (0.0087)	-0.0065 (0.0087)	-0.0055 (0.0087)
Board diversity	-0.2343*** (0.0899)	-0.2742*** (0.0886)	-0.2621*** (0.0886)	-0.2356*** (0.0897)
Board independence	1.2948*** (0.2344)	1.3069*** (0.2341)	1.3029*** (0.2339)	1.3044*** (0.2337)
Firm size	0.0913*** (0.0109)	0.095*** (0.0109)	0.0808*** (0.0110)	0.0861*** (0.0111)
Firm age	-0.0103*** (0.0022)	-0.0096*** (0.0022)	-0.0101*** (0.0022)	-0.0096*** (0.0022)
Firm leverage	-0.0420 (0.0279)	-0.0427 (0.0279)	-0.0433 (0.0278)	-0.0469* (0.0278)
Business growth	0.00004 (0.0010)	3.51e-06 (0.0010)	-0.00007 (0.0010)	-0.00003 (0.0010)
State ownership	-0.2305*** (0.0289)	-0.2135*** (0.0289)	-0.2437*** (0.0289)	-0.2335*** (0.0290)
Constant	-2.1895*** (0.2648)	-1.8482*** (0.2684)	-2.1697*** (0.2639)	-1.9569*** (0.2688)
Year fixed effects	Yes	Yes	Yes	Yes
Adj R-squared	0.1640	0.1664	0.1668	0.1708

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=11261

Table 4 Regressions of CFO characteristics on digital transformation (adding control variable)

Variables	Model 1	Model 2	Model 3	Model 4
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CFO gender	0.0631*** (0.0239)			0.0474** (0.0239)
CFO age		-0.0110*** (0.0018)		-0.0088*** (0.0018)
CFO education			0.1036*** (0.0154)	0.0836*** (0.0158)
CEO gender	0.0059 (0.0454)	0.0101 (0.0454)	0.0019 (0.0453)	0.0108 (0.0453)
CEO age	-0.0035** (0.0016)	-0.0022 (0.0016)	-0.0034** (0.0016)	-0.0025 (0.0016)
CEO education	0.0989*** (0.0122)	0.0989*** (0.0122)	0.0841*** (0.0124)	0.0853*** (0.0124)
Board size	-0.0088 (0.0087)	-0.0071 (0.0087)	-0.0078 (0.0087)	-0.0068 (0.0087)
Board diversity	-0.2262** (0.0899)	-0.2658*** (0.0887)	-0.2525*** (0.0887)	-0.2269** (0.0897)
Board independence	1.3127*** (0.2345)	1.3246*** (0.2342)	1.3229*** (0.2341)	1.3234*** (0.2338)
Firm size	0.0944*** (0.0110)	0.0983*** (0.0110)	0.0841*** (0.0111)	0.0892*** (0.0112)
Firm age	-0.0106*** (0.0022)	-0.0099*** (0.0022)	-0.0105*** (0.0022)	-0.0100*** (0.0022)
Firm leverage	-0.0448 (0.0279)	-0.0455 (0.0279)	-0.0465* (0.0279)	-0.0499* (0.0279)
Business growth	0.00007 (0.0010)	0.00003 (0.00103)	-0.00004 (0.0010)	-4.23e-06 (0.0010)
State ownership	-0.2241*** (0.0290)	-0.2074*** (0.0290)	-0.2369*** (0.0291)	-0.2271*** (0.0292)
Largest holder rate	-0.1684** (0.0801)	-0.1660** (0.0799)	-0.1869** (0.0799)	-0.1781*** (0.0799)
Constant	-2.1805*** (0.2648)	-1.8406*** (0.2683)	-2.1602*** (0.2639)	-1.9493*** (0.2688)
Year fixed effects	Yes	Yes	Yes	Yes
Adj R-squared	0.1643	0.1666	0.1671	0.1690

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=11261

Table 5 Regressions of CFO characteristics on digital transformation (using sub-sample 2007-2016)

Variables	Model 1	Model 2	Model 3	Model 4
CFO gender	0.1209***			0.0976***

	(0.0327)			(0.0328)
CFO age		-0.0122***		-0.0095***
		(0.0024)		(0.0024)
CFO education			0.1039***	0.0805***
			(0.0194)	(0.0199)
CEO gender	0.2464***	0.2461***	0.2374***	0.2483***
	(0.0652)	(0.0651)	(0.0651)	(0.0649)
CEO age	-0.0125***	-0.0107***	-0.0122***	-0.01124***
	(0.0023)	(0.0024)	(0.0023)	(0.0024)
CEO education	0.0831***	0.0836***	0.0681***	0.0648***
	(0.0168)	(0.0167)	(0.0171)	(0.0171)
Board size	-0.0057	-0.0029	-0.0045	-0.0031
	(0.0106)	(0.0106)	(0.0106)	(0.0106)
Board diversity	0.1338	0.0379	0.0614	0.1161
	(0.1304)	(0.1283)	(0.1283)	(0.1299)
Board independence	0.8650***	0.8799***	0.8435***	0.8736***
	(0.2992)	(0.2988)	(0.2987)	(0.2979)
Firm size	0.0311**	0.0369**	0.0213	0.0295**
	(0.0148)	(0.0148)	(0.0149)	(0.0149)
Firm age	-0.0059**	-0.0053*	-0.0055*	-0.0056*
	(0.0030)	(0.0030)	(0.0030)	(0.0030)
Firm leverage	-0.0699***	-0.0683***	-0.0709***	-0.0735***
	(0.0250)	(0.0249)	(0.0249)	(0.0249)
Business growth	-0.0023	-0.0032	-0.0027	-0.0034
	(0.0061)	(0.0061)	(0.0061)	(0.0061)
State ownership	-0.2230***	-0.2038***	-0.2384***	-0.2274***
	(0.0356)	(0.0356)	(0.0357)	(0.0358)
Constant	-0.6332*	-0.2654	-0.5759*	-0.4321
	(0.3381)	(0.3408)	(0.3366)	(0.3415)
Year fixed effects	Yes	Yes	Yes	Yes
Adj R-squared	0.1476	0.1500	0.1505	0.1549

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=4422

Table 6 Regressions of CFO characteristics on digital transformation (using sub-sample 2014-2023)

Variables	Model 1	Model 2	Model 3	Model 4
CFO gender	0.0687**			0.0552*
	(0.0283)			(0.0283)
CFO age		-0.0123***		-0.0101***
		(0.0022)		(0.0022)

CFO education			0.1111*** (0.0189)	0.0882*** (0.0194)
CEO gender	-0.0219 (0.0531)	-0.0154 (0.0529)	-0.0266 (0.0529)	-0.0151 (0.0529)
CEO age	-0.0026 (0.0019)	-0.0012 (0.0019)	-0.0025 (0.0019)	-0.0015 (0.0019)
CEO education	0.1176*** (0.0145)	0.1186*** (0.0145)	0.1033*** (0.0147)	0.1059*** (0.0147)
Board size	-0.0067 (0.0112)	-0.0055 (0.0112)	-0.0055 (0.0112)	-0.0047 (0.0112)
Board diversity	-0.3051*** (0.1046)	-0.3393*** (0.1033)	-0.3303*** (0.1033)	-0.2998*** (0.1043)
Board independence	1.4956*** (0.2932)	1.5017*** (0.2927)	1.5138*** (0.2927)	1.5065*** (0.2923)
Firm size	0.1130*** (0.0140)	0.1172*** (0.0140)	0.1004*** (0.0142)	0.1069*** (0.0142)
Firm age	-0.0110*** (0.0026)	-0.0102*** (0.0026)	-0.0109*** (0.0026)	-0.0103*** (0.0026)
Firm leverage	-0.0973 (0.0732)	-0.1019 (0.0731)	-0.0887 (0.0731)	-0.1029 (0.0731)
Business growth	0.00009 (0.0011)	0.00007 (0.0011)	-0.00002 (0.0011)	0.00003 (0.0011)
State ownership	-0.2639*** (0.0364)	-0.2448*** (0.0364)	-0.2767*** (0.0365)	-0.2663 (0.0366)
Constant	-2.1091*** (0.3336)	-1.7059*** (0.3384)	-2.1011*** (0.3326)	-1.8380*** (0.3389)
Year fixed effects	Yes	Yes	Yes	Yes
Adj R-squared	0.0637	0.0665	0.0667	0.0691

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=8893

Table 7 Regressions of CFO characteristics on digital transformation (Poisson regression)

Variables	Model 1	Model 2	Model 3	Model 4
CFO gender	0.0581*** (0.0072)			0.0432*** (0.0072)
CFO age		-0.01397*** (0.0005)		-0.0117*** (0.0006)

CFO education			0.12345*** (0.0048)	0.1015*** (0.0049)
CEO gender	0.0682*** (0.0139)	0.0755*** (0.0139)	0.0649*** (0.0139)	0.0751*** (0.0139)
CEO age	-0.0030*** (0.0005)	-0.0015*** (0.0005)	-0.0029*** (0.0005)	-0.0018*** (0.0005)
CEO education	0.1752*** (0.0037)	0.1752*** (0.0037)	0.1591*** (0.0038)	0.1614*** (0.0038)
Board size	-0.0577*** (0.0028)	-0.0558*** (0.0028)	-0.0561*** (0.0028)	-0.0546*** (0.0028)
Board diversity	-0.6582*** (0.0267)	-0.6765*** (0.0264)	-0.6674*** (0.0264)	-0.6377*** (0.0267)
Board independence	1.0496*** (0.0698)	1.0427*** (0.0699)	1.0757*** (0.0698)	1.0675*** (0.0699)
Firm size	0.1814*** (0.0034)	0.1861*** (0.0034)	0.1671*** (0.0034)	0.1742*** (0.0035)
Firm age	-0.0309*** (0.0007)	-0.0301*** (0.0007)	-0.0309*** (0.0007)	-0.0302*** (0.0007)
Firm leverage	-0.4165*** (0.0199)	-0.4202*** (0.0198)	-0.4017*** (0.0199)	-0.4165*** (0.0198)
Business growth	-0.0009** (0.0004)	-0.0009** (0.0004)	-0.0010*** (0.0004)	-0.0010*** (0.0004)
State ownership	-0.2881*** (0.0097)	-0.2633*** (0.0097)	-0.3031*** (0.0097)	-0.2864*** (0.0098)
Constant	-4.9404*** (0.1508)	-4.5155*** (0.1513)	-4.9188*** (0.1506)	-4.6419*** (0.1515)
Year fixed effects	Yes	Yes	Yes	Yes
Pseudo R ²	0.1446	0.1468	0.1469	0.1486

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=11261

Table 8 Regressions of CFO characteristics on digital transformation (two years lag)

Variables	Model 1	Model 2	Model 3	Model 4
CFO gender	0.0707*** (0.0265)			0.0548** (0.0266)
CFO age		-0.0114*** (0.0020)		-0.0093*** (0.0021)

CFO education			0.0975*** (0.0169)	0.0768*** (0.0173)
CEO gender	0.0717 (0.0509)	0.0750 (0.0509)	0.0688 (0.0508)	0.0769 (0.0509)
CEO age	-0.0035* (0.0018)	-0.0022 (0.0018)	-0.0034* (0.0018)	-0.0025 (0.0018)
CEO education	0.1085*** (0.0135)	0.1091*** (0.0135)	0.0956*** (0.0137)	0.0966*** (0.0137)
Board size	-0.0141 (0.0095)	-0.0123 (0.0095)	-0.0131 (0.0095)	-0.0118 (0.0095)
Board diversity	-0.1937* (0.1005)	-0.2385** (0.0990)	-0.2241** (0.0990)	-0.1946* (0.1002)
Board independence	1.1349*** (0.2623)	1.1571*** (0.2620)	1.1376*** (0.2619)	1.1527*** (0.2617)
Firm size	0.0846*** (0.0122)	0.0888*** (0.0122)	0.0748*** (0.0123)	0.0804*** (0.0124)
Firm age	-0.0102*** (0.0024)	-0.0095*** (0.0024)	-0.0100*** (0.0024)	-0.0096*** (0.0024)
Firm leverage	-0.0659* (0.0347)	-0.0669* (0.03464)	-0.0674* (0.0346)	-0.0716* (0.0346)
Business growth	0.0022 (0.0025)	0.0019 (0.0025)	0.0021 (0.0025)	0.0018 (0.0025)
State ownership	-0.2589*** (0.0319)	-0.2408*** (0.0318)	-0.2724*** (0.0319)	-0.2617*** (0.0321)
Constant	-1.9263*** (0.2945)	-1.5770*** (0.2983)	-1.9084*** (0.2936)	-1.6928*** (0.2989)
Year fixed effects	Yes	Yes	Yes	Yes
Adj R-squared	0.1490	0.1513	0.1514	0.1535

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=9351

Table 9 Regressions of CFO characteristics on digital transformation (GMM)

Variables	Model 1	Model 2	Model 3	Model 4
L. Digital transformation	0.5759*** (0.0544)	0.5753*** (0.0552)	0.5800*** (0.0585)	0.5793*** (0.0598)
CFO gender	0.0816 (0.4838)			0.1613 (0.5251)
CFO age		-0.0271 (0.0273)		-0.0204 (0.0306)

CFO education			0.4987 (0.3524)	0.4981 (0.3609)
CEO gender	0.2985 (0.8940)	0.4369 (0.9395)	0.0303 (0.9759)	0.1571 (1.0523)
CEO age	-0.0167 (0.0241)	-0.0161 (0.0249)	-0.0284 (0.0282)	-0.0273 (0.0285)
CEO education	0.4824** (0.2452)	0.4973** (0.2540)	0.4651* (0.2784)	0.4819* (0.2809)
Board size	-0.1915 (0.1454)	-0.1441 (0.1515)	-0.1558 (0.1561)	-0.1164 (0.1698)
Board diversity	-0.9258 (1.8147)	-0.7985 (1.7275)	-1.8638 (1.8311)	-1.8908 (2.0504)
Board independence	3.1461 (3.0877)	3.4494 (3.1590)	3.2517 (3.4863)	3.4999 (3.5564)
Firm size	0.7230*** (0.2643)	0.7089*** (0.2687)	0.7044** (0.2906)	0.6994** (0.3076)
Firm age	-0.1886* (0.1024)	-0.1894* (0.1035)	-0.1955* (0.1132)	-0.2008* (0.1190)
Firm leverage	-0.8358 (0.8876)	-0.8295 (0.9098)	-1.1059 (0.9790)	-1.0799 (0.9909)
Business growth	0.0032 (0.0253)	0.0009 (0.0259)	-0.0049 (0.0293)	-0.0067 (0.0302)
State ownership	0.5973 (0.7789)	0.8408 (0.8617)	0.9039 (0.8670)	1.0751 (0.9509)
Constant	-11.8840** (5.5583)	-11.0343* (5.8352)	-12.1045* (6.2778)	-11.5224* (6.6563)
Year fixed effects	Yes	Yes	Yes	Yes
AR(1)	0.0000	0.0000	0.0000	0.0000
AR(2)	0.685	0.322	0.579	0.924
Hansen test	0.939	0.790	0.924	0.977

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses. Number of Observations=11261

Table 10 Analysis based on firm size

Variables	Model 1		Model 2		Model 3	
	Larger firms	Smaller firms	Larger firms	Smaller firms	Larger firms	Smaller firms
CFO gender	0.1158*** (0.0329)	0.0015 (0.0346)				
CFO age			-0.0098***	-0.0122***		

			(0.0025)	(0.0025)		
CFO education					0.0820*** (0.0211)	0.1153*** (0.0225)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	6232	5029	6232	5029	6232	5029
Adj R-squared	0.1489	0.1728	0.1492	0.1767	0.1493	0.1771
Differences (p values)		0.0169		0.5081		0.2779

Note: ***p<0.01, **p<0.05, *p<0.1. Standard errors are in parentheses.

Table 11 Analysis based on firm age

Variables	Model 1		Model 2		Model 3	
	Older firms	Younger firms	Older firms	Younger firms	Older firms	Younger firms
CFO gender	0.0396 (0.0320)	0.0919*** (0.0356)				
CFO age			-0.0078*** (0.0024)	-0.0149*** (0.00267)		
CFO education					0.0818*** (0.0212)	0.1199*** (0.0221)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	6561	4700	6561	4700	6561	4700
Adj R-squared	0.1012	0.2258	0.1025	0.2299	0.1031	0.2296
Differences (p values)		0.2776		0.0448		0.2086

Note: ***p<0.01, **p<0.05, *p<0.1. Standard errors are in parentheses.

Table 12 Analysis based on ownership type

Variables	Model 1		Model 2		Model 3	
	SOE = 1	SOE = 0	SOE = 1	SOE = 0	SOE = 1	SOE = 0
CFO gender	0.1231*** (0.0459)	0.0497* (0.0279)				
CFO age			-0.0029 (0.0034)	-0.0132*** (0.0021)		
CFO education					-0.0170	0.1454***

					(0.0282)	(0.0182)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	3011	8250	3011	8250	3011	8250
Adj R-squared	0.2295	0.1349	0.2278	0.1387	0.2277	0.1412
Differences (p values)	0.1677		0.0130		0.0000	

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are in parentheses.